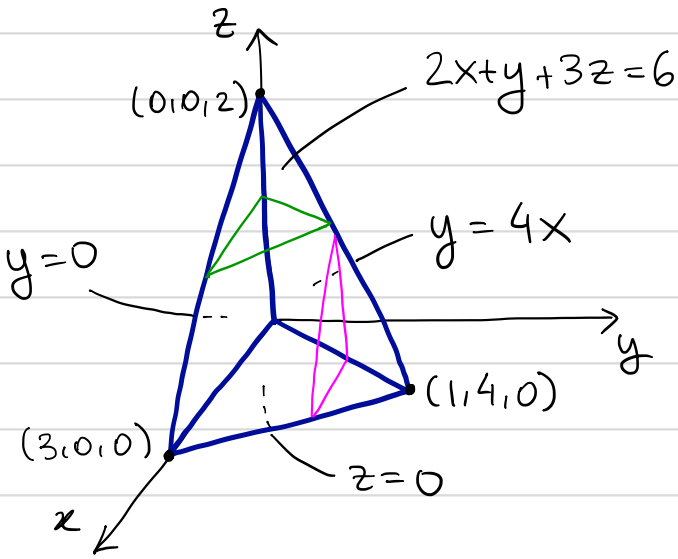
Lecture 6

Triple integrals (15.6)

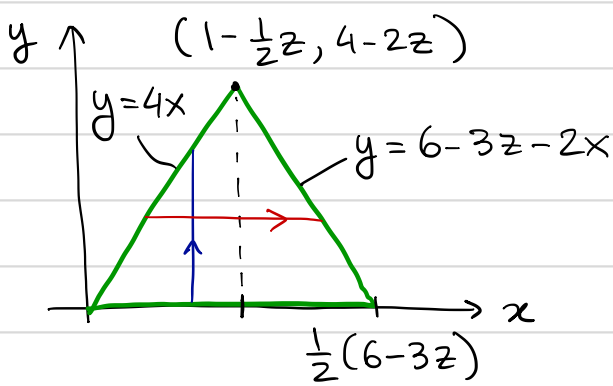
Cylindrical coordinates (15.7)

Example: E = tetrahedron



$$\iiint_E f(x,y,z)\,dV =$$

$$= \int_{z=0}^{2} \int_{y=0}^{4-2z} \int_{x=\frac{1}{4}y}^{\frac{1}{2}(6-y-3z)} f \, dx\,dy\,dz$$



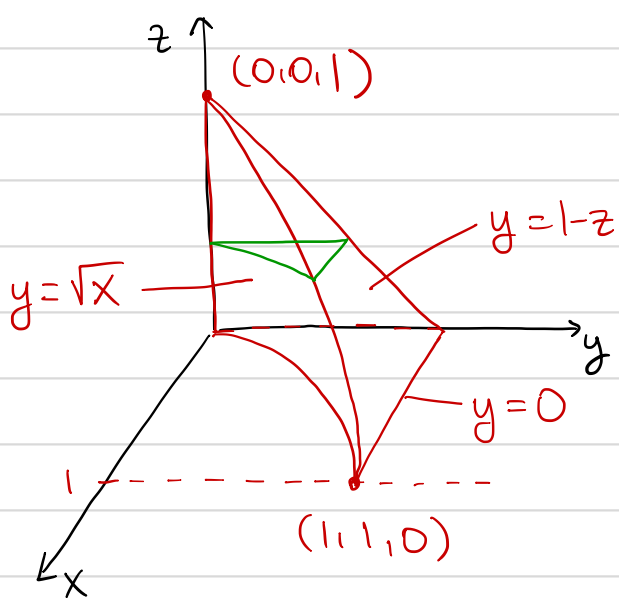
$$= \int_{z=0}^{2} \left[\int_{x=0}^{1-\frac{1}{2}z} \int_{y=0}^{4x} f \, dy\,dx + \int_{x=1-\frac{1}{2}z}^{\frac{1}{2}(6-3z)} \int_{y=0}^{6-3z-2x} f \, dy\,dx \right] dz$$

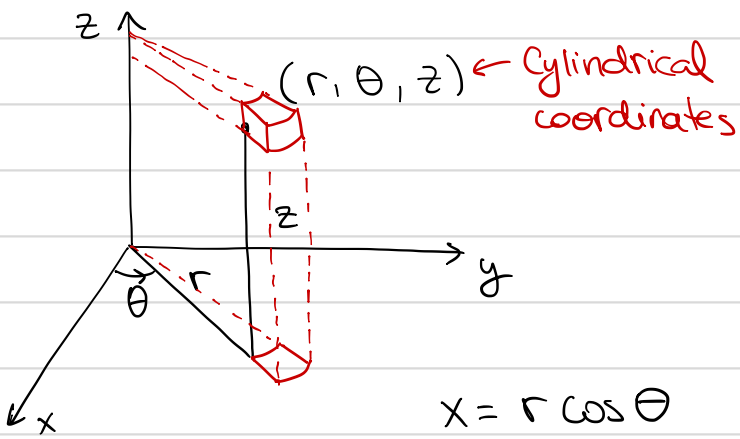
$$= \int_{y=0}^{4} \int_{x=\frac{1}{4}y}^{\frac{1}{2}(6-y)} \int_{z=0}^{\frac{1}{3}(6-2x-y)} f \, dz\,dx\,dy$$

Example:

$$\int_{x=0}^{1} \int_{y=\sqrt{x}}^{1} \int_{z=0}^{1-y} f(x,y,z)\,dz\,dy\,dx =$$

$$= \int_{z=0}^{1} \int_{y=0}^{1-z} \int_{x=0}^{y^2} f(x,y,z)\,dx\,dy\,dz$$





$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$

Volume element: $dV = r\,dr\,d\theta\,dz$

Mass $= \iiint_E \rho\, dV$

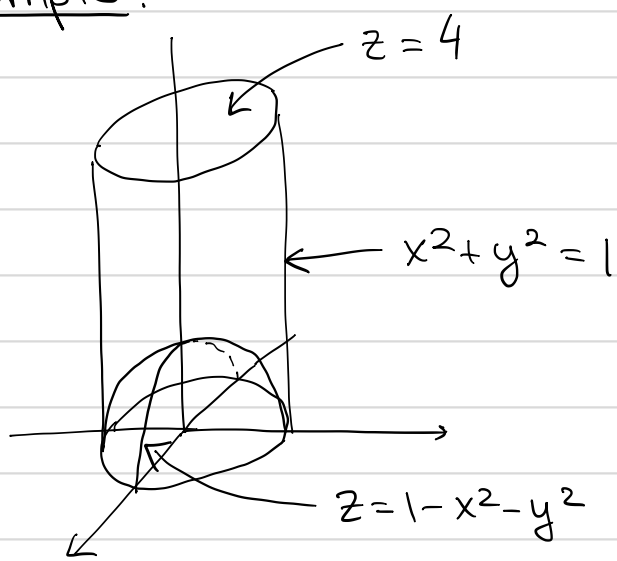
$$= \int_0^{2\pi}\int_0^1\int_{z=1-r^2}^{4} Kz\,dz\,r\,dr\,d\theta$$

$$= 2\pi K \int_0^1 \underbrace{\frac{1}{2}\left(16-(1-2r^2+r^4)\right)}_{\frac{15}{2}+r^2-\frac{1}{2}r^4} r\,dr$$

$$= 2\pi K \left[\frac{15}{4}r^2 + \frac{r^3}{3} - \frac{1}{10}r^5\right]_{r=0}^{1}$$

$$= 2\pi K\left(\frac{15}{4}+\frac{1}{3}-\frac{1}{10}\right) //$$

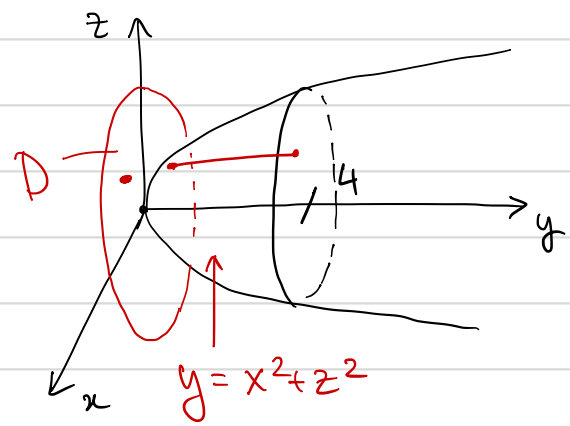
Example:

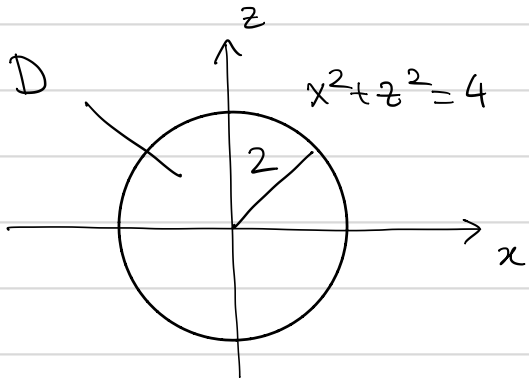


Density: $\rho = Kz$

Example: $\iiint_E \sqrt{x^2+z^2}\,dV$

E = region between $y = x^2+z^2$ and $y = 4$





$$\iiint_E \sqrt{x^2+z^2}\, dV =$$

$x = r\cos\theta$

$z = r\sin\theta$

$y = y$

$$= \iint_D \int_{y=r^2}^{4} r \, dy \, dA$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{2} (4-r^2)\, r \cdot r \, dr \, d\theta$$

$$= 2\pi \left[\frac{4r^3}{3} - \frac{r^5}{5} \right]_0^2 = 2\pi \left[\frac{32}{3} - \frac{32}{5} \right]$$

$$= \frac{128}{15}\pi \quad //$$